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        REM ORIGINAL MATRIX DESTROYED.                                    UAINV1 001
        REM FLOATING POINT.                                                 INV1 002
        REM MATRIX INVERSION USING CORE STORAGE ONLY.                       INV1 003
        REM COEFFICIENTS OF ORIGINAL MATRIX IN ROW ORDER.                   INV1 004
        REM INVERSE IN SAME LOCATION AS ORIGINAL MATRIX.                    INV1 005
        REM STORAGE REQUIREMENTS.                                           INV1 006
        REM    MATRIX, LA11 TO LA11&N SQUARE -1                             INV1 007
        REM    PROGRAM, INV TO INV&170                                      INV1 008
        REM    COMMON, COMMON TO COMMON&2N&7                                INV1 009
        REM CALLING SEQUENCE.                                               INV1 010
        REM    TSX INV,4                                                    INV1 011
        REM    HTR LA11,0,N                                                 INV1 012
        REM    RETURN, SINGULAR SET                                         INV1 013
        REM    RETURN, NORMAL                                               INV1 014
        REM                                                                 INV1 015
        REM                                                                 INV1 016
INV     DCT                    TURN OFF DIV. CHK. INDICATOR                 INV1 017
  1     NOP                                                                 INV1 018
  2     SXD COMMON&1,1         SAVE INDEX 1                                 INV1 019
  3     SXD COMMON&2,2         SAVE INDEX 2                                 INV1 020
  4     PXD                    CLEAR TO 0                                   INV1 021
  5     STO COMMON                                                          INV1 022
  6     STO COMMON&3                                                        INV1 023
  7     STO COMMON&4                                                        INV1 024
  8     CLA 1,4                OBTAIN N                                     INV1 025
  9     STD COMMON&3                                                        INV1 026
 10     STD INV&67                                                          INV1 027
 11     STD INV&112                                                         INV1 028
 12     STD INV&155                                                         INV1 029
 13     ARS 18                                                              INV1 030
 14     STO COMMON&5                                                        INV1 031
 15     ADD INV&78             FIRST LOCATION OF SECOND HALF OF ERASE       INV1 032
 16     STA INV&34                                                          INV1 033
 17     STA INV&79                                                          INV1 034
 18     STA INV&80                                                          INV1 035
 19     STA INV&118                                                         INV1 036
 20     STA INV&119                                                         INV1 037
 21     STA INV&132                                                         INV1 038
 22     STA INV&136                                                         INV1 039
 23     STA INV&137                                                         INV1 040
 24     SUB INV&169            LAST LOCATION OF FIRST HALF OF ERASE         INV1 041
 25     STA INV&121                                                         INV1 042
 26     ADD COMMON&5           LAST LOCATION OF ERASE                       INV1 043
 27     STA INV&94                                                          INV1 044
 28     STA INV&98                                                          INV1 045
 29     ADD INV&169            LAST LOCATION OF ERASE&1                     INV1 046
 30     STA INV&103                                                         INV1 047
 31     STA INV&115                                                         INV1 048
 32     CLA INV&169            SET UP ROW IDENTIFICATION IN FIRST HALF      INV1 049
 33     LXD COMMON&3,1            OF ERASE                                 INV1 050
 34     STO --,1                                                            INV1 051
 35     ADD INV&169                                                         INV1 052
 36     TIX INV&34,1,1                                                      INV1 053
 37     CLA 1,4                                                             INV1 054
 38     ADD COMMON&5           FIRST LOCATION OF SECOND ROW OF MATRIX       INV1 055
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39      STA INV&86                                                      INV1 056
40      LDQ COMMON&3        N SQUARE                                    INV1 057
41      MPY COMMON&3                                                    INV1 058
42      ALS 17                                                          INV1 059
43      STO COMMON&6                                                    INV1 060
44      STO COMMON&7                                                    INV1 061
45      ARS 18                                                          INV1 062
46      ADD 1,4                                                         INV1 063
47      ADD COMMON&5        LAST LOCATION OF MATRIX&N&1                 INV1 064
48      STA INV&100                                                     INV1 065
49      STA INV&108                                                     INV1 066
50      CLA COMMON&6        N SQUARE                                    INV1 067
        ARS 18                                                          INV1 068
        ADD 1,4             LAST LOCATION OF MATRIX                     INV1 069
        SUB INV&169                                                     INV1 070
        STA INV&109                                                     INV1 071
        STA INV&111                                                     INV1 072
        ADD INV&169         LAST LOCATION OF MATRIX&1                   INV1 073
        STA INV&116                                                     INV1 074
        STA INV&66                                                      INV1 075
        STA INV&68                                                      INV1 076
        STA INV&83                                                      INV1 077
        STA INV&87                                                      INV1 078
        STA INV&90                                                      INV1 079
        LXD COMMON&6,1      NUMBER OF NON-REDUCED ELEMENTS              INV1 080
        SXD COMMON,1                                                    INV1 081
        CLM                 FIND LARGEST LEADING ROW ELEMENT            INV1 082
        ADM --,1            LEADING ELEMENT FIRST ROW                   INV1 083
        TNX INV&72,1,--                                                 INV1 084
        LDQ --,1            LEADING ELEMENT ITH ROW                     INV1 085
        LRS 0                                                           INV1 086
        TLQ INV&67                                                      INV1 087
        TRA INV&64                                                      INV1 088
        CLA COMMON          INTERCHANGE ROW IDENTIFICATION IN ERASE     INV1 089
        LRS 35                                                          INV1 090
        DVH COMMON&3                                                    INV1 091
        LLS 35                                                          INV1 092
        ADD COMMON&4        LOCATION REDUCTION NUMBER-1                 INV1 093
        PAX 0,1                                                         INV1 094
        CLA COMMON&8        FIRST LOCATION OF FIRST HALF OF ERASE       INV1 095
        LDQ --,1            FIRST LOCATION OF SECOND HALF OF ERASE      INV1 096
        STO --,1                                                        INV1 097
        STQ COMMON&8                                                    INV1 098
        LXD COMMON,1                                                    INV1 099
        CLA --,1                                                        INV1 100
        STO COMMON          DIVISOR                                     INV1 101
        LXD COMMON&3,2                                                  INV1 102
        CLA --,2            MOVE LEADING ROW                            INV1 103
        STO --,1                                                        INV1 104
        TNX INV&96,2,1                                                  INV1 105
        TNX INV&90,1,1                                                  INV1 106
        CLA --,1                                                        INV1 107
        FDP COMMON          FORM PIVOTAL ROW                            INV1 108
        DCT                 TEST FOR SINGULAR SYSTEM                    INV1 109
        TRA INV&162                                                     INV1 110
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STQ --,2                                                        INV1 111
TRA INV&86                                                      INV1 112
CLA INV&170                                                     INV1 113
FDH COMMON          FORM 1/DIVISOR                              INV1 114
STQ --                                                          INV1 115
LXD COMMON&7,1                                                  INV1 116
CLA --,1                                                        INV1 117
STO COMMON          MULTIPLIER                                  INV1 118
LXD COMMON&3,2                                                  INV1 119
LDQ --,2                                                        INV1 120
FMP COMMON          MULTIPLY PIVOTAL ROW BY LEADING ELEMENT     INV1 121
CHS                    OF ROW I AND SUBTRACT                    INV1 122
TNX INV&107,1,1                                                 INV1 123
TNX INV&111,2,1                                                 INV1 124
FAD --,1                                                        INV1 125
STO --,1                                                        INV1 126
TRA INV&103                                                     INV1 127
STO --,1                                                        INV1 128
TXH INV&100,1,--                                                INV1 129
LDQ COMMON&8                                                    INV1 130
LXD COMMON&3,2                                                  INV1 131
CLA --,1            MOVE PIVOTAL ROW TO MATRIX FROM             INV1 132
STO --,1               SECOND HALF OF ERASE                     INV1 133
TNX INV&118,2,1                                                 INV1 134
CLA --,2            ROTATE ROW IDENTIFICATION IN FIRST HALF     INV1 135
STO --,1               OF ERASE                                 INV1 136
TIX INV&115,1,1                                                 INV1 137
STQ --                                                          INV1 138
CLA COMMON&6                                                    INV1 139
SUB COMMON&3                                                    INV1 140
STO COMMON                                                      INV1 141
STO COMMON&6                                                    INV1 142
TNZ INV&163                                                     INV1 143
CLA INV&169                                                     INV1 144
LXD COMMON&3,1                                                  INV1 145
STO COMMON                                                      INV1 146
PXD 0,1                                                         INV1 147
PDX 0,2                                                         INV1 148
CLA --,2            TEST FOR COLUMN ORDER                       INV1 149
SUB COMMON                                                      INV1 150
TZE INV&136                                                     INV1 151
TIX INV&132,2,1                                                 INV1 152
CLA --,1            INTERCHANGE COLUMN IDENTIFICATION           INV1 153
STO --,2                                                        INV1 154
PXD 0,1             SET UP ADDRESSES FOR INTERCHANGING COLUMNS   INV1 155
SUB COMMON&3                                                    INV1 156
ARS 18                                                          INV1 157
SUB INV&116                                                     INV1 158
STA INV&151                                                     INV1 159
STA INV&154                                                     INV1 160
PXD 0,2                                                         INV1 161
SUB COMMON&3                                                    INV1 162
ARS 18                                                          INV1 163
SUB INV&116                                                     INV1 164
STA INV&152                                                     INV1 165
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STA  INV&153
LXD  COMMON&7,2
CLA  --,2              INTERCHANGE COLUMNS
LDQ  --,2
STO  --,2
STQ  --,2
TIX  INV&151,2,--
CLA  COMMON
ADD  INV&169
TIX  INV&129,1,1
LXD  COMMON&1,1        RESTORE INDEX 1
LXD  COMMON&2,2        RESTORE INDEX 2
TRA  3,4               NORMAL EXIT
TXI  INV&159,4,1       SINGULAR SET EXIT
CLA  COMMON&4          INCREASE REDUCTION NUMBER
ADD  INV&169
STO  COMMON&4
CLA  INV&66            MODIFY ADDRESSES FOR NEXT REDUCTION
SUB  COMMON&5
TRA  INV&58
DEC  1                 FIXED CONSTANT 1
DEC  1.                FLOATING CONSTANT 1
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INV1 166
INV1 167
INV1 168
INV1 169
INV1 170
INV1 171
INV1 172
INV1 173
INV1 174
INV1 175
INV1 176
INV1 177
INV1 178
INV1 179
INV1 180
INV1 18
INV1 18
INV1 18
INV1 18
INV1 18
INV1 18
INV1 18
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